

REWORK IN HIGHWAY PROJECTS: A TEST PAPER

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Abstract. There is limited systematic knowledge available about the dynamics of rework in highway projects, despite the fact that they frequently exceed budget and schedule by more than 10%. A case study of a highway project, which experienced a significant cost overrun as a result of rework, is examined and the causal factors that contributed to it are determined.

Keywords: highway projects, rework, system dynamics.

1. Introduction

The highway transport sector represents a large share of public spending. Cost overruns in transportation infrastructure projects, such as highways, have been identified as being attributable to errors and the subsequent rework that often occurs during construction works.

2. Research approach

With this in mind, the causal nature of rework that arose in eight highway projects is examined.

The findings are used to develop an influence diagram, based on the concept of system dynamics, to determine the interrelationships between variables that contributed to rework as well as time and cost overruns experienced. The developed model will enhance understanding about rework.

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